



SPRING 2026

Instructor: Dr. Rahul Bhadani

Deep Learning for Engineering Applications

Examination 1 Sample

February 26, 2026 • 4:20 PM - 5:40 PM • 100 Points

⚠ Important Instructions

- **Permitted:** One two-sided handwritten cheat sheets (letter size or smaller), calculator, pen/pencil, eraser
- **Prohibited:** textbooks, electronic devices, collaboration of any kind
- Read every question carefully; do not assume prior familiarity with phrasing
- Use all provided space; request extra sheets if needed
- Write clearly – illegible work receives **0 pts**
- **Circle** or **box** every final answer
- Show all work for partial credit on numerical sections

Name: _____

A#: _____

Total Pages: 24 • By signing below you affirm adherence to the academic integrity policy.

Signature: _____

The University of Alabama in Huntsville (UAH)

Section 1 – ☒ – True / False

For each statement, circle **T** (True) or **F** (False). Each question is worth **1 pts**. No partial credit.

1. The function

$$g(z) = \begin{cases} 1, & z \geq \theta \\ 0, & \text{otherwise} \end{cases}$$

is fully differentiable.

☐ T ☐ F

2. Is it possible to implement a neural network in PyTorch without inheriting from `nn.Module`?

☐ T ☐ F

3. The vanishing gradient problem becomes *more* severe as network depth increases, particularly with saturating activations such as sigmoid and tanh.

☐ T ☐ F

4. In a Convolutional Neural Network, filter weights are shared spatially across all locations of a given feature map.

☐ T ☐ F

5. The Adam optimizer maintains per-parameter adaptive learning rates by tracking exponentially decaying first and second moment estimates of the gradient.

☐ T ☐ F

6. Predicting the glucose level given patients' health metrics is a regression problem.

☐ T ☐ F

7. I have got the training accuracy of 95% while training a deep learning model for predicting WiFi strength given access point transmit power, antenna gain/direction, and WiFi channel/frequency. I should immediately declare victory and publish a paper based on the result.

☐ T ☐ F

8. `uv venv - python 3.12` creates a Python virtual environment with Python 3.12.

☐ T ☐ F

9. The Coefficient of determination R^2 can also be applied to deep neural network-based machine learning models.

☐ T ☐ F

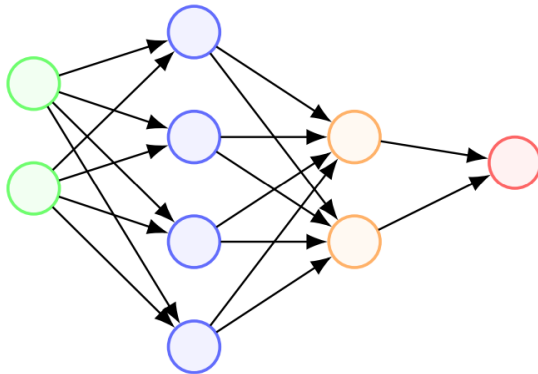
10. Sigmoid function is not differentiable at least at one point.

☐ T ☐ F

Section 2 – Multiple Choice

Circle the single best answer. **1 pts each.** No partial credit.

Q1. Consider a Neural Network diagram:



If we are supplying two samples with two features (green bubbles), then what is the dimension of W s responsible for compressing 4 meta features to 2 meta features? For this, assume that features in a feature matrix are along the column dimensions, and samples are along row dimensions.

- (A) $W_{4 \times 2}$ (B) $W_{2 \times 4}$ (C) $W_{4 \times 4}$ (D) $W_{2 \times 2}$

Q2. A 2-D input of size 228×228 is passed through a convolutional layer with 64 filters of size 3×3 , stride 1, and padding 1. What is the **spatial dimension** of the output feature map?

- (A) 226×226 (B) 230×230 (C) 228×228 (D) 224×224

Q3. Consider a Network definition

```
1 class Network(nn.Module):
2     def __init__(self):
3         super().__init__()
4         self.layer1 = nn.Linear(10, 20)
5         self.layer2 = nn.Linear(20, 5)
6     def forward(self, x):
7         return self.layer2(self.layer1(x))
```

How many trainable parameters are there in the above network (including biases)?

(A) 325

(B) 300

(C) 55

(D) 400

Q4. Which of the following is not an assumption in the Simple Linear Regression model?

(A)

$$\mathbb{E}[\epsilon_i] = 0 \quad \forall i$$

(B)

$$\sigma^2[\epsilon_i] = \sigma^2 \text{ (a constant)} \quad \forall i$$

(C)

$$\sigma[\epsilon_i, \epsilon_j] = 0 \quad \forall i \neq j$$

(D)

$$\mathbb{E}[\epsilon_i^2] = 0 \quad \forall i$$

Q5. Which of the loss functions is not suitable for regression problems?

- (a) Cross-Entropy Loss
- (b) Mean-squared Error Loss
- (c) Mean absolute Error Loss
- (d) Total squared loss

Q6. Consider an activation function as $\sigma(\mathbf{x}) = \frac{1}{1+e^{-\mathbf{x}}}$, if input is $\mathbf{x} = [1.09, 2.01, 3.0]$, the what is the dimension of $A = \sigma(\mathbf{x})$?

- (a) 3×3
- (b) 1
- (c) Cannot be determined
- (d) 3

Q7. Consider a Neural Network definition:

```

1  class QuizModel(nn.Module):
2      def __init__(self):
3          super().__init__()
4          self.fc1 = nn.Linear(10, 5)
5          self.relu = nn.ReLU()
6          self.fc2 = nn.Linear(5, 2)
7
8      def forward(self, x):
9          x = self.fc1(x)
10         x = self.relu(x)
11         x = self.fc2(x)
12         return x

```

What is the size of the final output?

- (a) 2
- (b) 5
- (c) 1
- (d) 2×5

Q8. Consider

$$\mathbf{x} = \begin{bmatrix} 1 \\ 2 \\ 4 \\ 6 \end{bmatrix}$$

and

$$\mathbf{y} = \mathbf{f}(\mathbf{x}) = \begin{bmatrix} x_1 x_2 \\ x_1 + x_2 \\ x_1 x_2 + x_2 x_3 \end{bmatrix}$$

What is the size of the Jacobian J ?

- (a) 4×4
- (b) 4×3
- (c) 3×4
- (d) 3×3

Q9. Consider a convolution layer definition

```
1 conv = nn.Conv2d(in_channels=4,  
2                   out_channels=6,  
3                   kernel_size=3,  
4                   stride=2,  
5                   padding=1)
```

What is the dimension of an example tensor that can serve as an input to the above convolution layer?

- (a) `torch.Size([2, 1, 5, 6])`
- (b) `torch.Size([2, 6, 32, 32])`
- (c) `torch.Size([2, 3, 5, 6])`
- (d) `torch.Size([2, 4, 32, 32])`

Q10. Applying max pooling with a kernel size of 3×3 to the image of size 5×5 with a stride of 2, the resulting output has size

- (a) 2×2
- (b) 3×3
- (c) 5×5
- (d) 3×5

Section 3 – – Short Answer

Answer each prompt in the box. Write in complete sentences. **5 pts each.**

3.1

Explain the intuition behind the *attention mechanism* in transformers. Why is it more expressive than a fixed recurrent hidden state? **[5 pts]**

Answer

3.2

What happens if you don't register a member variable tensor as parameters in PyTorch `nn.Module`? **[5 pts]**

Answer

3.3

Why does a feedforward fully connected neural network not work well with sequential data? **[5 pts]**

Answer

3.4

What is the Markov assumption in an n-Gram Language Model?

[5 pts]**Answer**

Section 4 – – Mathematically Trained**4.1**

An input feature matrix is

$$I = \begin{bmatrix} 3 & 3 & 2 & 2 & 1 & 1 \\ 4 & 4 & 4 & 0 & 0 & 0 \\ 7 & 6 & 0 & 0 & 0 & 0 \\ 0 & 0 & 3 & 5 & 0 & 0 \end{bmatrix}$$

Consider an averaging filter Kernel

$$K = \frac{1}{9} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

Consider a padding of 1 and a stride of 1, and compute the output after applying the kernel to the input feature matrix.

4.2

Section 5 – – Forward Pass

Show all work for partial credit. Box or circle every final numeric answer.

Problem 4.1**[5 pts]** Two-layer fully connected network:

- Input: $\mathbf{x} = [1.0, 0.5]^T$
- Layer 1: $W^{(1)} = \begin{bmatrix} 0.2 & -0.3 \\ 0.5 & 0.8 \end{bmatrix}$, $\mathbf{b}^{(1)} = [0.1, -0.2]^T$, activation: **ReLU**
- Layer 2: $W^{(2)} = [0.6, -0.4]$, $b^{(2)} = 0.0$, activation: **Sigmoid**

(a) Compute $\mathbf{z}^{(1)} = W^{(1)}\mathbf{x} + \mathbf{b}^{(1)}$.

[1 pts]

(b) Apply ReLU to obtain $\mathbf{a}^{(1)} = \text{ReLU}(\mathbf{z}^{(1)})$.

[1 pts]

(c) Compute $\hat{y} = \sigma(W^{(2)}\mathbf{a}^{(1)} + b^{(2)})$.

[2 pts]

(d) If $y = 1$, compute the binary cross-entropy loss.

[1 pts]

Section 6 – – Magical CNN

A CNN processes 3-channel 224×224 images through:

Layer	Filters	Kernel	Stride	Padding
Conv1	32	3×3	1	1
Conv2	64	3×3	1	1
MaxPool	–	2×2	2	0
Conv3	128	3×3	1	1

1.

(a) How many trainable parameters are there in **Conv1**?

[2 pts]

(b) What is the spatial resolution after **MaxPool**?

[2 pts]

(c) How many trainable parameters are there in **Conv3**?

[2 pts]

Section 7 – – Code Identification Identify the Architecture (10 pts)

Examine the PyTorch snippet below and answer the questions that follow.

```

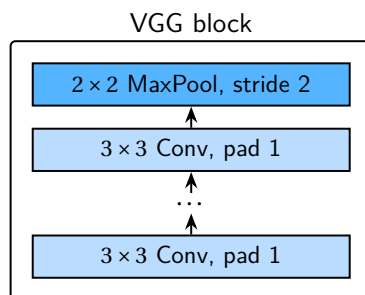
1  import torch
2  import torch.nn as nn
3
4  class MysteryBlock(nn.Module):
5      def __init__(self, in_channels, out_channels):
6          super().__init__()
7          self.branch1 = nn.Sequential(
8              nn.Conv2d(in_channels, out_channels, kernel_size=1),
9              nn.BatchNorm2d(out_channels),
10             nn.ReLU(inplace=True),
11             nn.Conv2d(out_channels, out_channels, kernel_size=3, padding=1),
12             nn.BatchNorm2d(out_channels),
13         )
14         self.downsample = (
15             nn.Conv2d(in_channels, out_channels, kernel_size=1)
16             if in_channels != out_channels else nn.Identity()
17         )
18         self.relu = nn.ReLU(inplace=True)
19
20     def forward(self, x):
21         identity = self.downsample(x)
22         out = self.branch1(x)
23         out = out + identity          # <<< Line A
24         return self.relu(out)

```

1. What well-known deep learning architecture does `MysteryBlock` implement? **[2 pts]**
2. What is the purpose of **Line A** and the `self.downsample` layer? Explain in 2–3 sentences. **[4 pts]**
3. What specific training problem does this design help solve? Name the problem and state how this construction addresses it mathematically (hint: consider gradient flow). **[4 pts]**

Section 8 – – Fancy Layers**8.1**

Consider a VGG Block:



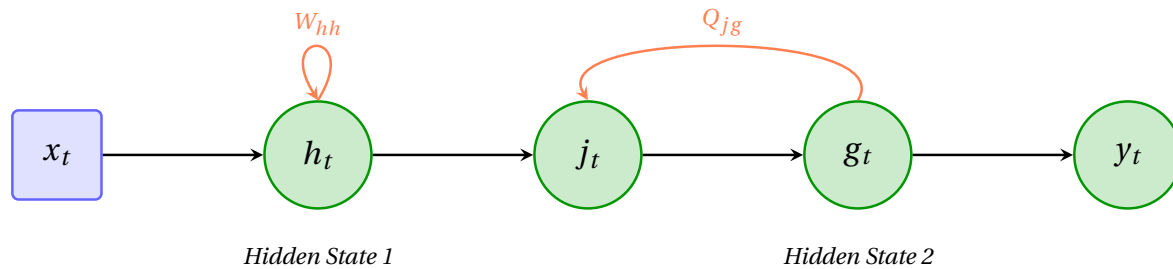
Provide an implementation for a composite layer `VGGBlock` representing the above VGG Block implemented in PyTorch by inheriting from `nn.Module` class that specifies v copies of a 3×3 Conv Layer of padding 1 sequentially connected.

8.2

Consider a quadratic equation $h(\mathbf{x}) = a\mathbf{x}^2 + b\mathbf{x} + c$. Provide an implementation for a custom Neural Network Layer called QuadraticNet with a , b , and c as trainable parameters using PyTorch by inheriting from `nn.Module`.

8.3

Consider a network shown below.



Provide an implementation of the above neural network with a recurrent connection and a feedback using PyTorch by inheriting from `nn.Module`. Assume that all nodes are performing linear affine transformation, and in addition to recurrence and feedback, they also apply linear affine transformation to regular inputs.

Section 9 – – Sentence Probability

Consider a text corpus:

```
1      <s> Washington had hoped to make a stand at New Brunswick, but was  
2 disappointed</s>  
3      <s> New Brunswick is a province of Canada</s>  
4      <s> Washington has never been to Canada</s>
```

What is the probability of

```
1      <s>Washington had hoped to make a stand at Canada</s>
```

using a 2-gram model?

Section 10 – – Embeddings

Consider that after a particular epoch during the training embedding matrix looks like

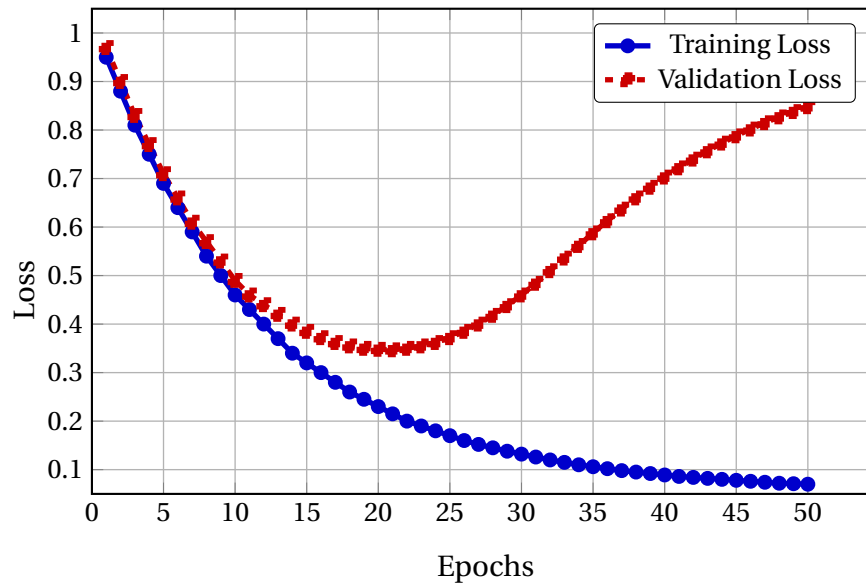
$$E = \begin{bmatrix} 0.12 & 0.47 & 0.83 & 0.26 & 0.59 & 0.34 \\ 0.71 & 0.05 & 0.62 & 0.48 & 0.91 & 0.17 \\ 0.33 & 0.88 & 0.14 & 0.76 & 0.42 & 0.65 \\ 0.56 & 0.29 & 0.73 & 0.08 & 0.84 & 0.51 \\ 0.97 & 0.43 & 0.31 & 0.67 & 0.22 & 0.78 \\ 0.15 & 0.64 & 0.52 & 0.39 & 0.86 & 0.03 \\ 0.68 & 0.21 & 0.95 & 0.57 & 0.13 & 0.44 \\ 0.82 & 0.36 & 0.07 & 0.93 & 0.61 & 0.28 \end{bmatrix}$$

Answer the following question:

1. What is the d_{model} of the embedding?
2. What is the size of the vocabulary?
3. Write the one-hot encoding matrix corresponding to the vocabulary.
4. Assuming that sequences ID assigned to tokens in the vocabulary start from 1, if a particular token sequence has been represented by $T = \{4, 2, 1, 5, 3, 4\}$, what is the embedding for the given sequence?

Section 11 – 🎮 – Training Master

Training vs. Validation Loss



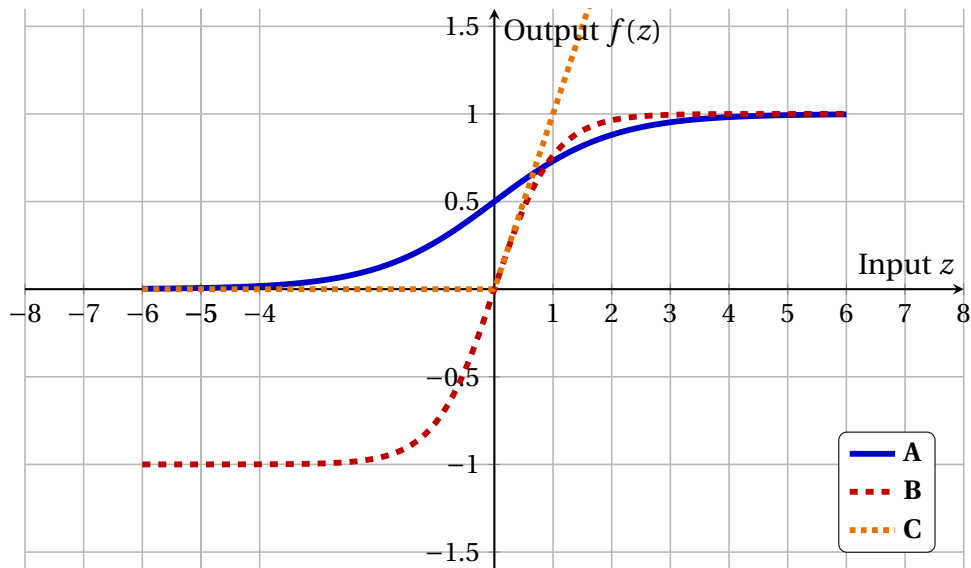
1.

By looking at the above graph, when would have been the best time to stop training the neural network??

2. **Activation Functions**

Identify the different activation functions.

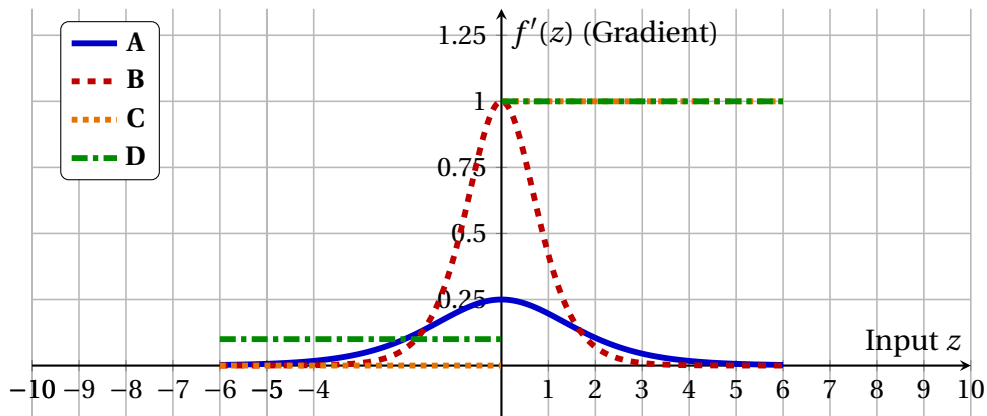
Which curve suffers from saturation or dying neurons?



3. Derivatives of Activation Functions

Identify different activation functions based on their derivatives.

Where does the gradient vanish or die?



Section 12 – – Code Writing – Implement SimpleCNN (10 pts)

Complete the *SimpleCNN* class in PyTorch matching the specification below. Input images are 28×28 grayscale.

- **Conv1:** 1 ch in, 32 filters, 3×3 , pad 1, ReLU
- **MaxPool:** 2×2
- **Conv2:** 32 ch in, 64 filters, 3×3 , pad 1, ReLU
- **MaxPool:** 2×2
- **FC1:** flattened $\rightarrow$ 128 hidden, ReLU
- **Dropout:** $p = 0.5$
- **FC2:** 128 $\rightarrow$ 10 logits (no softmax)

```
1 import torch
2 import torch.nn as nn
3 import torch.nn.functional as F
4
5 class SimpleCNN(nn.Module):
6     def __init__(self):
7         super().__init__()
8         # Define all layers below
9
10
11
12
13
14
15
16
17
18
19         # end __init__
20
21     def forward(self, x):
22         # Forward pass below
23
24
25
26
27
28
29
30         # end forward
31         return x
```

 **Bonus (+2 pts)**

Write a single Python expression that counts all trainable parameters of an instantiated SimpleCNN model.

If you need to implement weight initialization for Conv1, how would you do it, and when should the weight initialization function be called?

Section 13 – – Jacobian and Derivatives

1. Consider $y = \sum_{i=1}^n f_i(\mathbf{x}, \mathbf{z})$ where $f_i(\mathbf{x}, \mathbf{z}) = x_i^2 z_i$. $\mathbf{x}, \mathbf{z}$ each has n elements.
 - (a) Compute the gradient $\nabla_{\mathbf{x}} y$. What's its dimension?
 - (b) Compute the gradient $\nabla_{\mathbf{z}} y$. What's its dimension?
2. Consider $y = f(x) = e^{x^2-1}$. Introduce intermediate variables, compute intermediate derivatives, and apply the chain rule to determine $\frac{dy}{dx}$.
3. Find the Jacobian of

$$f(x, y, z) = \begin{bmatrix} ye^z \\ x - y \\ 3y + z^2 \end{bmatrix}$$

 End of Examination

Please double-check your name on Page 1.